

Exercise 1

Assume that $y_1, \dots, y_{200}$ are realizations of independent Gaussian random variables with variance equal to 1 and mean $\beta_1 + \beta_2 \exp\{z_i\}$ for $i = 1, \dots, 120$, and mean $\beta_1 + \beta_3 \exp\{z_i^2\}$ for $i = 121, \dots, 200$; where the z_i are known constants and $(\beta_1, \beta_2, \beta_3)$ are unknown real parameters.

- a) Are the assumptions of a Gaussian linear model satisfied in the above formulation? Motivate the answer.
- b) State the parameter space and sample space.
- c) Express the model in matrix form: $\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, explicitly stating how $\underline{Y}$, X , $\underline{\beta}$, and $\underline{\varepsilon}$ are defined and their dimensions. Write the distribution of $\underline{Y}$ and $\underline{\varepsilon}$.
- d) Obtain the expression of the matrix $X^T X$ and the vector $X^T \underline{y}$; state how these elements should be used to obtain the maximum likelihood estimate $\hat{\underline{\beta}}$.
- e) Write the distribution of the maximum likelihood estimator $\hat{\underline{\beta}}(\underline{Y})$.
- f) Let $\underline{e} = \underline{y} - X\hat{\underline{\beta}}$ be the vector of the residuals. State which of the following identities are satisfied and motivate the answer:

$$\sum_{i=1}^{200} e_i = 0$$

$$\sum_{i=1}^{200} e_i z_i = 0$$

$$\sum_{i=1}^{200} e_i z_i^2 = 0$$

$$\sum_{i=1}^{200} e_i \exp\{z_i\} = 0$$

$$\sum_{i=1}^{200} e_i \exp\{z_i^2\} = 0$$

$$\sum_{i=1}^{120} e_i \exp\{z_i\} = 0$$

(hint: read the indices in the sum!)

$$y_1, \dots, y_{200}$$

- $i = 1, \dots, 120$
- $i = 121, \dots, 200$

~~$$y_i \sim \mathcal{N}(y_i) = \beta_1 + \beta_2 e^{z_i}$$~~~~$$y_i \sim \mathcal{N}(y_i) = \beta_1 + \beta_3 e^{z_i^2}$$~~

$$y_i \sim \mathcal{N}(\beta_1 + \beta_2 e^{z_i}; 1)$$
$$y_i \sim \mathcal{N}(\beta_1 + \beta_3 e^{z_i^2}; 1)$$

5) ASSUMPTIONS OF GAUSSIAN LINEAR MODEL SATISFIED?

YES THE ASSUMPTIONS ARE SATISFIED

- MODEL IS LINEAR IN $\beta_1, \beta_2, \beta_3 \rightarrow$ ARE LINEAR IN THE MEAN EXPRESSION
- NORMALITY $\rightarrow y_i$ FOLLOW GAUSSIAN RANDOM VARIABLES SO ERRORS ARE NORMALLY DISTRIBUTED
- HOMOGENEITY $\rightarrow$ VARIANCE = 1 EACH OBSERVATION y_i COMES FROM A DISTRIBUTION THAT HAS THE SAME VARIANCE (1)
- INDEPENDENCE $\rightarrow$ THE TEXT SAYS THAT y_i ARE REALIZATION OF INDEPENDENT GAUSSIAN RANDOM VARIABLES

6) PARAMETER SPACE - SAMPLE SPACE

- PARAM. SPACE: $\theta = \mathbb{R}^p, (A^T = 1 \times p, 1 = \mathbb{R}^3)$
- SAMPLE SPACE: $y = \mathbb{R}^n = \mathbb{R}^{200}$

$$\begin{bmatrix} y \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_{200} \end{bmatrix}$$

$$\underline{y} \sim \mathcal{N}_{200}(X\underline{\beta}, I_{200})$$
$$\underline{\varepsilon} \sim \mathcal{N}_{200}(0; I)$$

$$y_i = \beta_1 + \beta_2 x_{i,1} + \beta_3 x_{i,3} + \varepsilon_i$$

$$x_{i,2} = \begin{cases} e^{z_i} & i = 1, 2, \dots, 120 \\ 0 & \text{otherwise} \end{cases}$$
$$x_{i,3} = \begin{cases} e^{z_i^2} & i = 121, \dots, 200 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{bmatrix} 1 & x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & e^{z_1} & 0 & 0 \\ 1 & e^{z_2} & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & e^{z_{120}} & 0 & 0 \\ 1 & 0 & e^{z_{121}^2} & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & e^{z_{200}^2} & 0 \end{bmatrix}$$
$$y = \beta_1 + \beta_2 e^{z_i}$$

$$X^T X = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} 1 & x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1^T 1 & 1^T x_1 & 1^T x_2 & 1^T x_3 \\ x_1^T 1 & x_1^T x_1 & x_1^T x_2 & x_1^T x_3 \\ x_2^T 1 & x_2^T x_1 & x_2^T x_2 & x_2^T x_3 \\ x_3^T 1 & x_3^T x_1 & x_3^T x_2 & x_3^T x_3 \end{bmatrix} = \begin{bmatrix} 200 & \sum_{i=1}^{120} e^{z_i} & \sum_{i=121}^{200} e^{z_i^2} \\ \sum_{i=1}^{120} e^{z_i} & \sum_{i=1}^{120} e^{2z_i} & 0 \\ \sum_{i=121}^{200} e^{z_i^2} & 0 & \sum_{i=121}^{200} e^{2z_i^2} \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1^T \\ x_1^T \\ x_2^T \\ x_3^T \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_{200} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{200} y_i \\ \sum_{i=1}^{120} e^{z_i} y_i \\ \sum_{i=121}^{200} e^{z_i^2} y_i \end{bmatrix}$$

e DISTRIBUTION of $\hat{\beta}(y)$

$$\hat{\beta}(y) \sim N_p(\beta; \sigma^2(x^*x)^{-1})$$

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$$\sum_{i=1}^{200} e_i = 0 \quad \sum_{i=1}^{200} e_i z_i = 0 \quad \sum_{i=1}^{200} e_i z_i^2 = 0$$
$$\sum_{i=1}^{200} e_i \exp\{z_i\} = 0 \quad \sum_{i=1}^{200} e_i \exp\{z_i^2\} = 0 \quad \sum_{i=1}^{120} e_i \exp\{z_i\} = 0$$

1 $\sum_{i=1}^{200} e_i = 0$ HOLDS OTHERS NO